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Edge-Based Transportation Classifier for Real-Time CO_2 Estimation

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ECE-499 - Winter 2026
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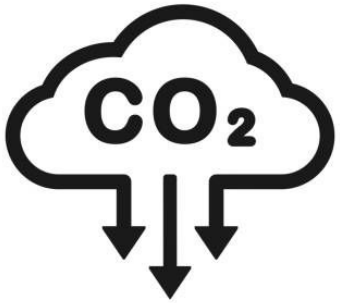
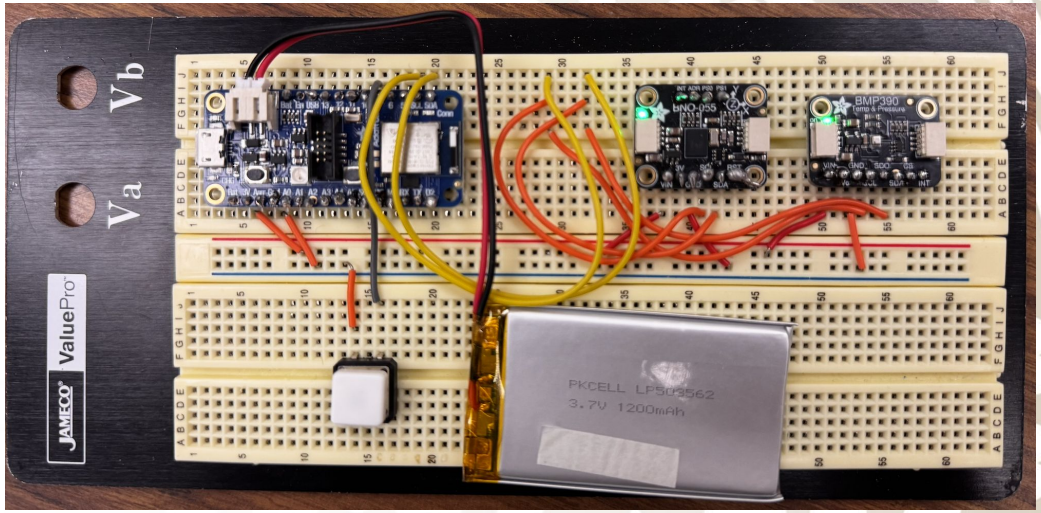
Project Recap

- **Goal:** Classify transportation modes using on-device sensor data
- **Target modes:** train, subway, car, bus, walk
  - Predicted mode feeds directly into CO_2 estimation
- **Constraints:**
  - Fully local data processing
  - Battery-powered embedded system

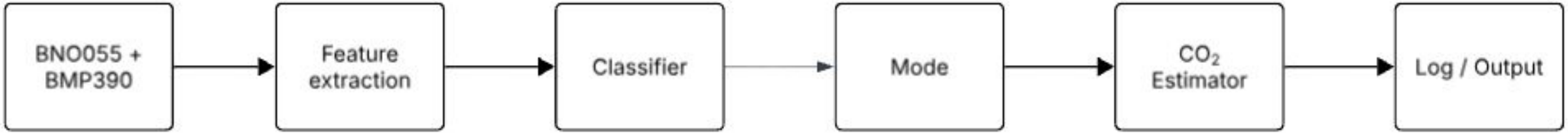


Overall System Design

- BNO055 (accelerometer + gyroscope)
- BMP390 (barometer)
- Feather nRF52840 (microcontroller)



Data flow:



Summary of Work Accomplished Since Week 3

Summary of Data Collection

- **Self-collected dataset only (no SHL yet)**
- **Total Data Collected:**
  - Train: 81 minutes (~121,000 samples)
  - Subway: 21 minutes (~31,500 samples)
  - Car: 42 minutes (~63,000 samples)
  - Bus: 58 minutes (~87,000 samples)
  - Walk: 16 minutes (~24,000 samples)

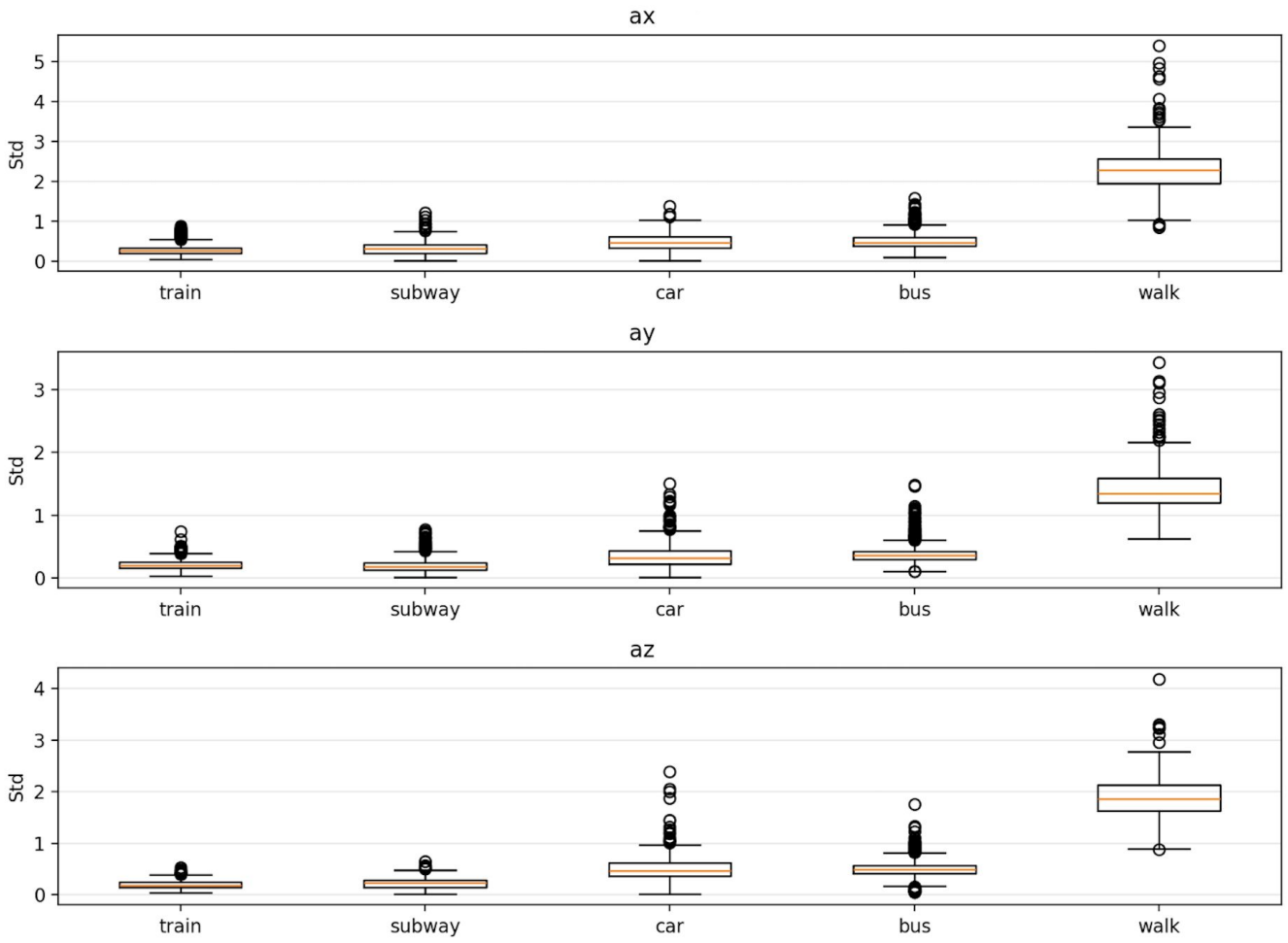
Data Preprocessing & Windowing

- **Windowing:** 5-second windows @ 25 hz (125 samples/window)
- **Channels/windows:** ax, ay, az, gx, gy, gz, pressure (7 total)
- **Normalization:** per-channel z-score using training-set mean/std

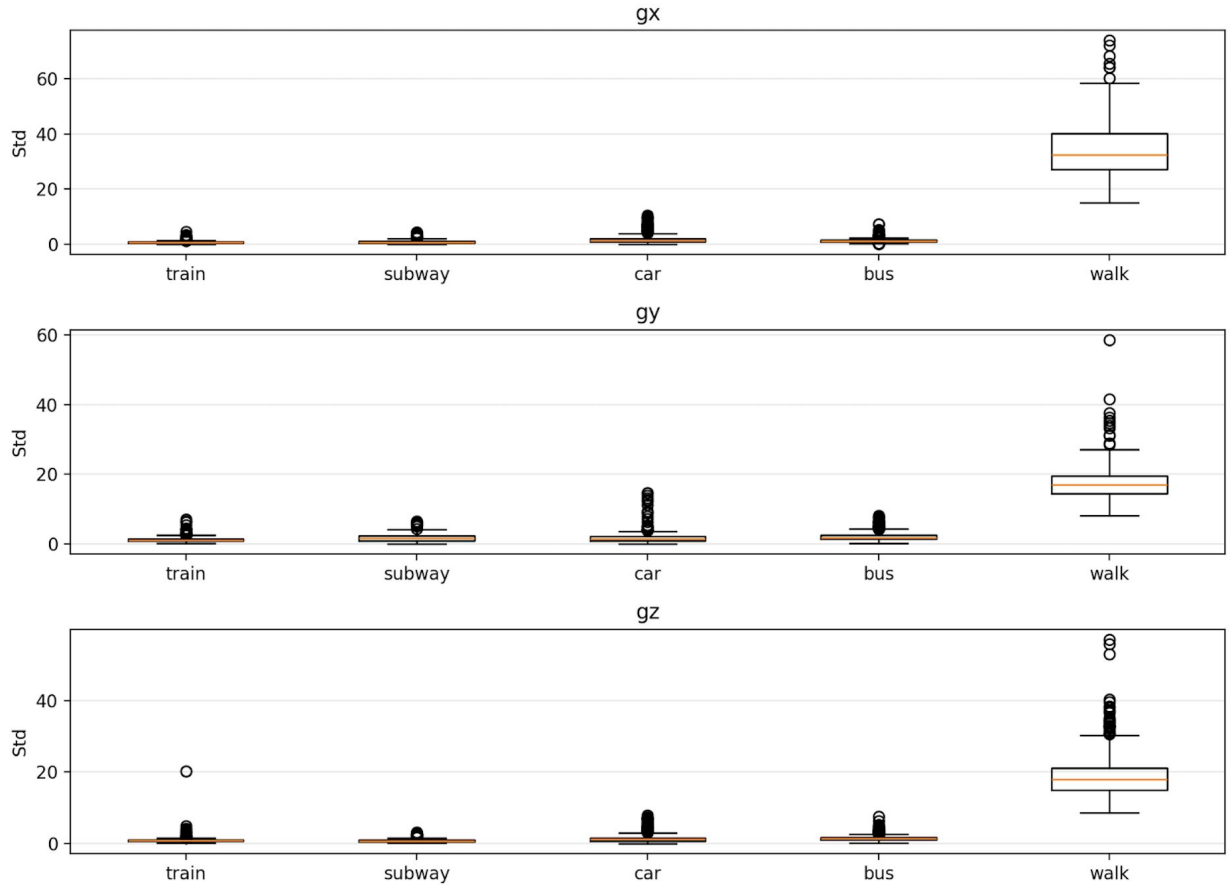
mode	test_windows	train_windows	n_test_files	n_train_files
train	985	903	2	3
subway	162	329	3	3
car	240	228	1	1
bus	424	971	1	2
walk	223	164	1	2

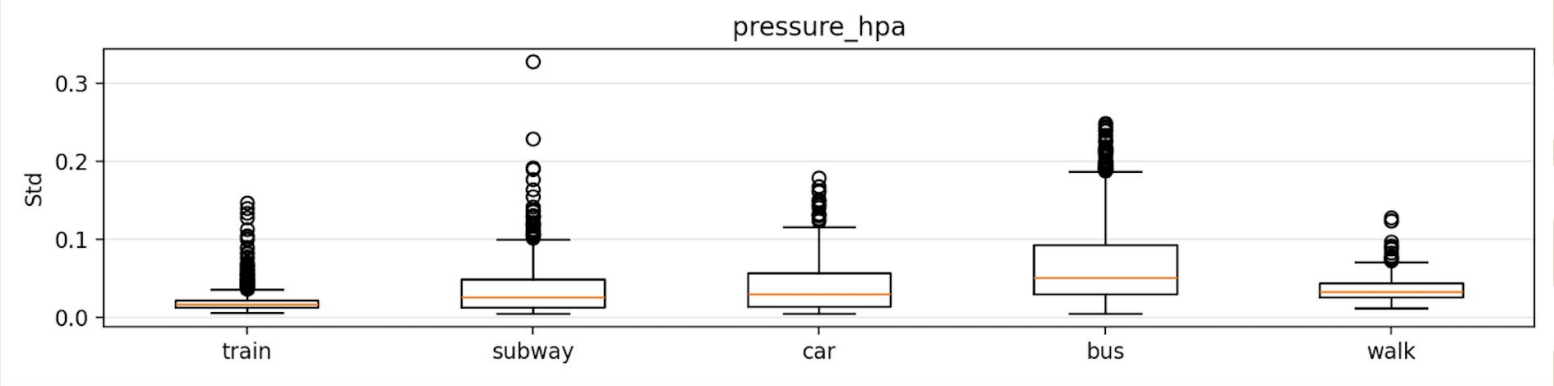
Feature Variability Analysis

- Standard deviation computed per 5 second window.
- Captures short-term motion dynamics
- Higher std -> more vibration/variation in that sensor during the window.



Walk is consistently separable, but the bus/train overlap indicates confusion risk.





Baseline Model Selection

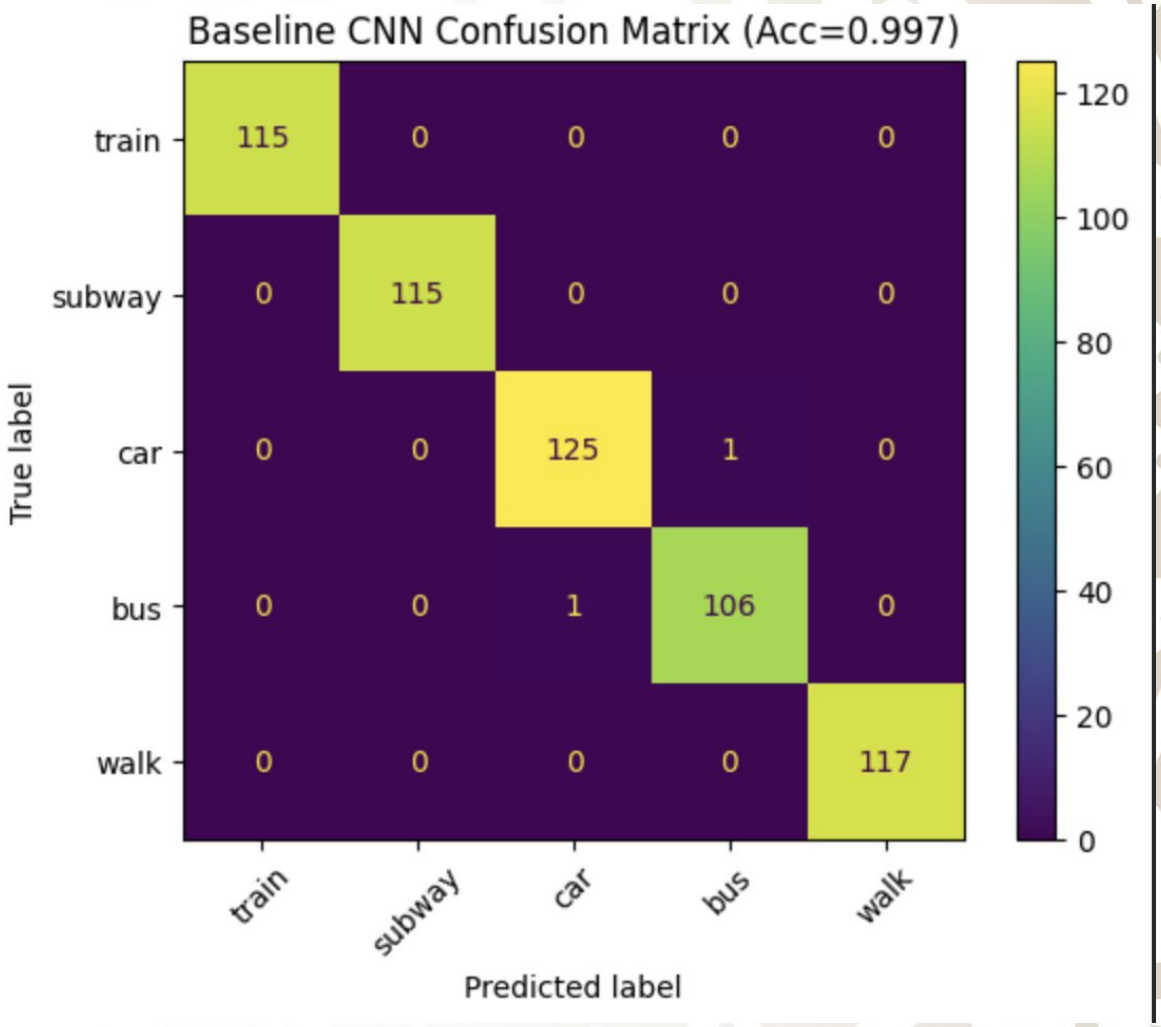
- **Input: 5-second window @ 25 Hz (125 samples x 7 channels)**
- **Model:** Convolutional Neural Network
  - Learns motion patterns automatically
  - Strong baseline for IMU time-series in the literature
- **Baseline goal:** establish a realistic generalization benchmark before adding SHL pretraining or extra features.

```
# model
model = tf.keras.Sequential([
    tf.keras.layers.Input(shape=(X_train.shape[1], X_train.shape[2], 1)),
    tf.keras.layers.Conv2D(16, (5,1), activation="relu", padding="same"),
    tf.keras.layers.MaxPool2D((2,1)),
    tf.keras.layers.Conv2D(32, (5,1), activation="relu", padding="same"),
    tf.keras.layers.MaxPool2D((2,1)),
    tf.keras.layers.Flatten(),
    tf.keras.layers.Dense(64, activation="relu"),
    tf.keras.layers.Dropout(0.3),
    tf.keras.layers.Dense(len(modes), activation="softmax")
])
```

Baseline Neural Network (Initial Result)

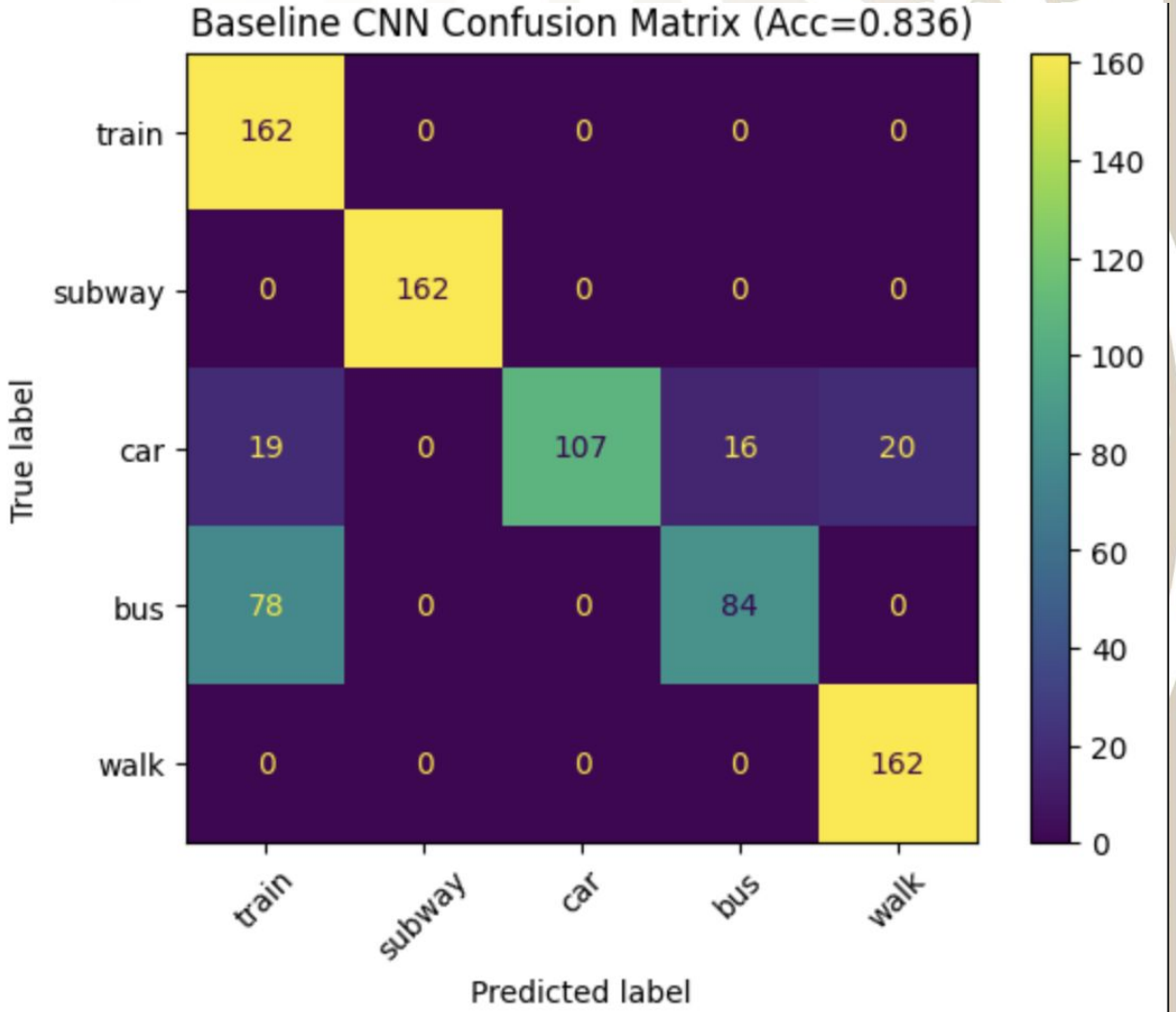
Initial accuracy: 99.7%

- Window-level random split caused leakage
- Not a valid estimate of generalization



Fixing Data Leakage

- **Fixed split:**
  - Entire segments assigned to train or test
  - No overlapping windows across split
- **Bus often misclassified with train, car also mixes with train/bus**
- **Baseline is limited to a small number of unique segments (bus/car)**



Classification Accuracy Target: ≥ 90%

Plan for Remaining Project

Timeframe	Focus	Key Tasks	Outcome
Week 7	Data and Model Generalization	• SHL preprocessing and pretraining • Fine-tune on self-collected data	Improved generalization on self-collected data
Week 8	Model Refinement	• Reduce bus/train confusion • Validate on held-out self segments	Stable confusion matrix
Week 9	CO_2 Estimation	• Apply emission factors per mode • Compare predicted vs. true CO_2 • Final Presentation	CO_2 error characterization
Weeks 10-11	Deployment & Finalization	• Model compression • Latency + power evaluation • Final Report	Final validated system

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Thank you

Questions?

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Appendix - SHL Dataset (Sussex - Huawei Locomotion)

SHL Dataset Signal	How It Is Used Used In My Project
Accelerometer (x, y, z)	Primary motion input for transportation classification
Gyroscope (x, y, z)	Captures rotational dynamics (turns, vibration patterns)
Barometric Pressure	Detects elevation changes and road profile
Linear Acceleration	Used for cleaner motion features
Orientation / Euler Angles	Provides posture and device orientation context
GPS (latitude, longitude, speed)	Ground-truth labeling and trip segmentation
Labeled Transportation Mode	Supervised learning target for classifier training
Multiple Body Positions	Evaluates robustness to sensor placement

Hips_Motion.txt

```
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```

Appendix - CO_2 Estimation Model

Objective: Estimate per-trip carbon emissions based on detected transportation mode, duration, and average speed, without continuous GPS tracking.

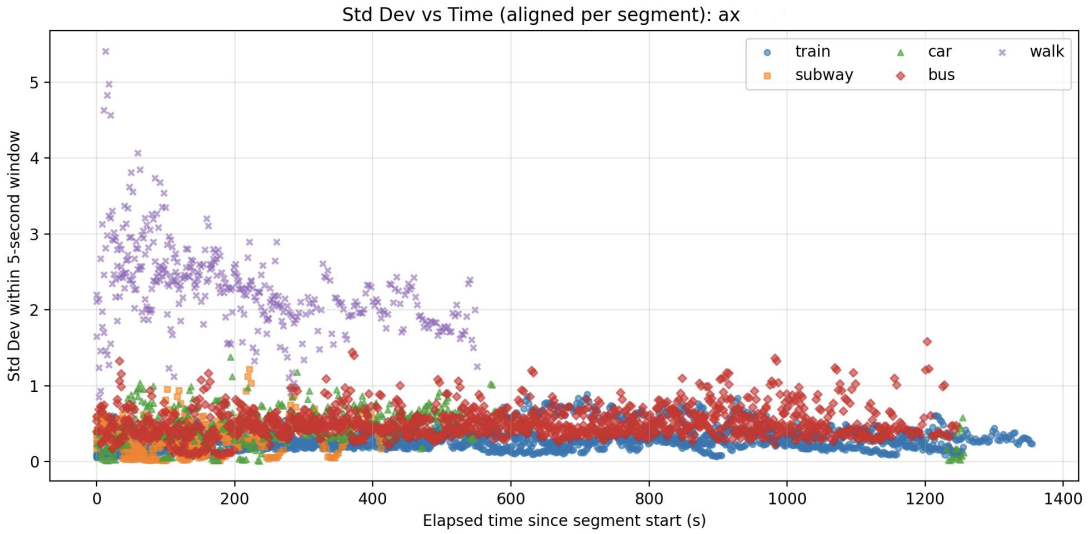
Calculation Framework:

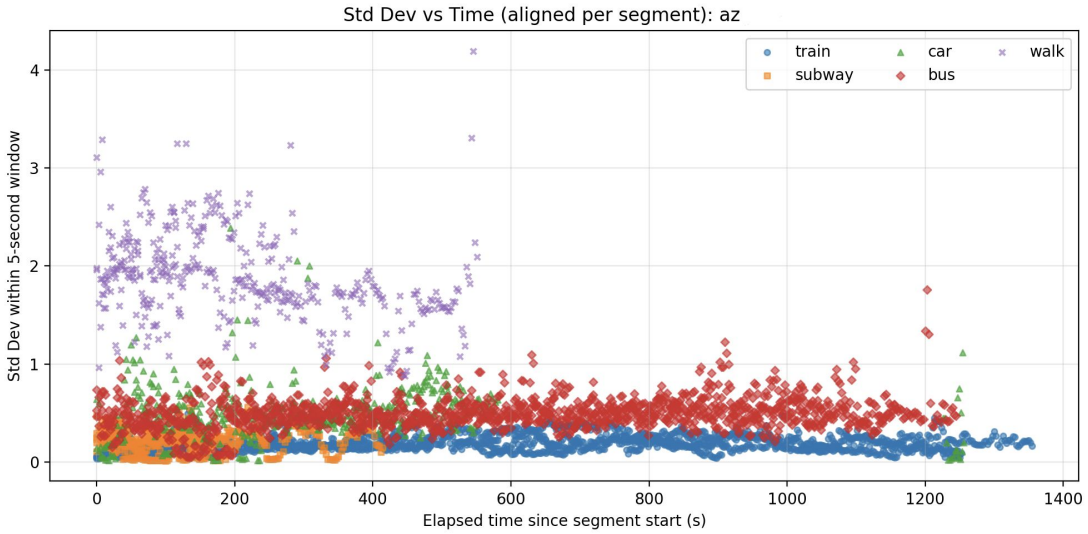
$$CO_{2trip} = EF_{mode} * D_{est}$$

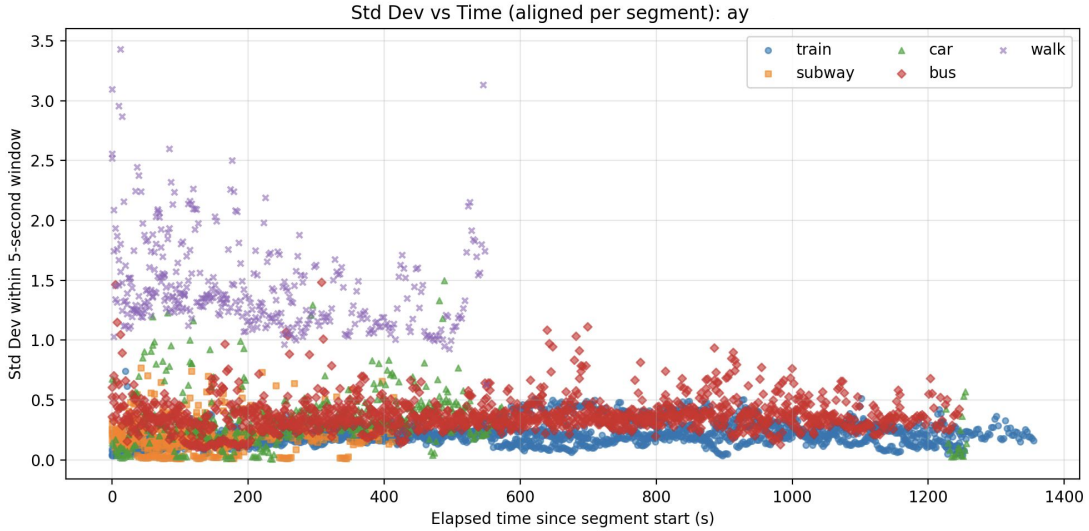
- EF_{mode} = Emission Factor (grams of CO_2 per km)
- D_{est} = Estimate Distance (km)
  - $D_{est} = v_{avg(mode)} * t_{mode}$

Mode	Emission Factor (g CO_2 / km)
Car	170 - 250 g CO_2 / km (per passenger)
Bus	60 - 110 g CO_2 / km (per passenger)
Train	25 - 45 g CO_2 / km (per passenger)
Subway	44 - 65 g CO_2 / km (per passenger)
Walk	0 g CO_2 / km (per passenger)

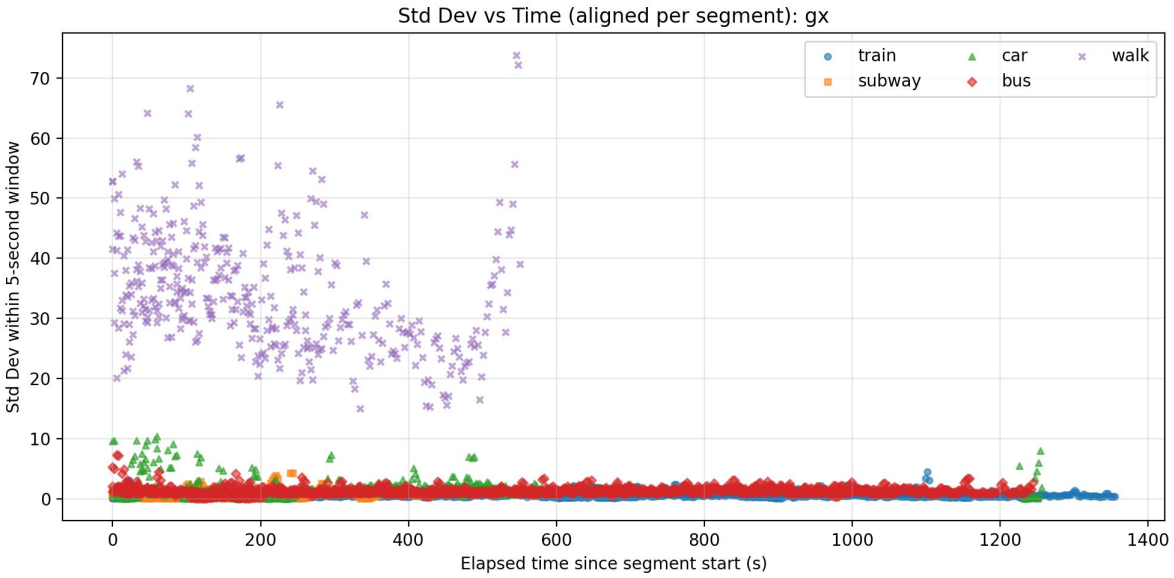
Appendix - Acceleration Graphs

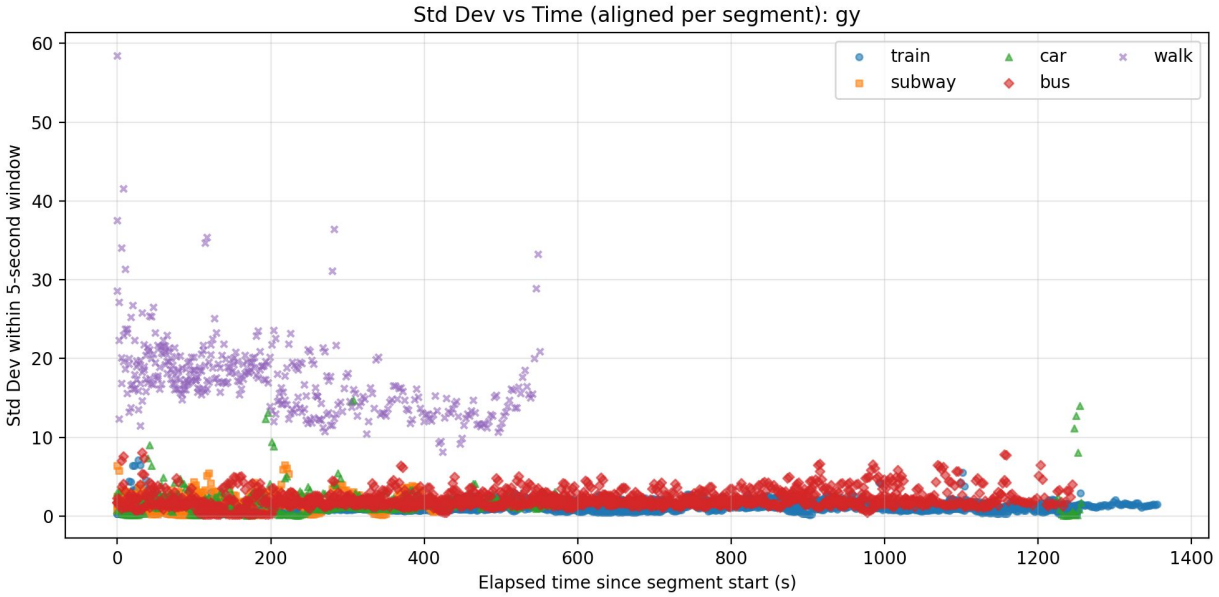


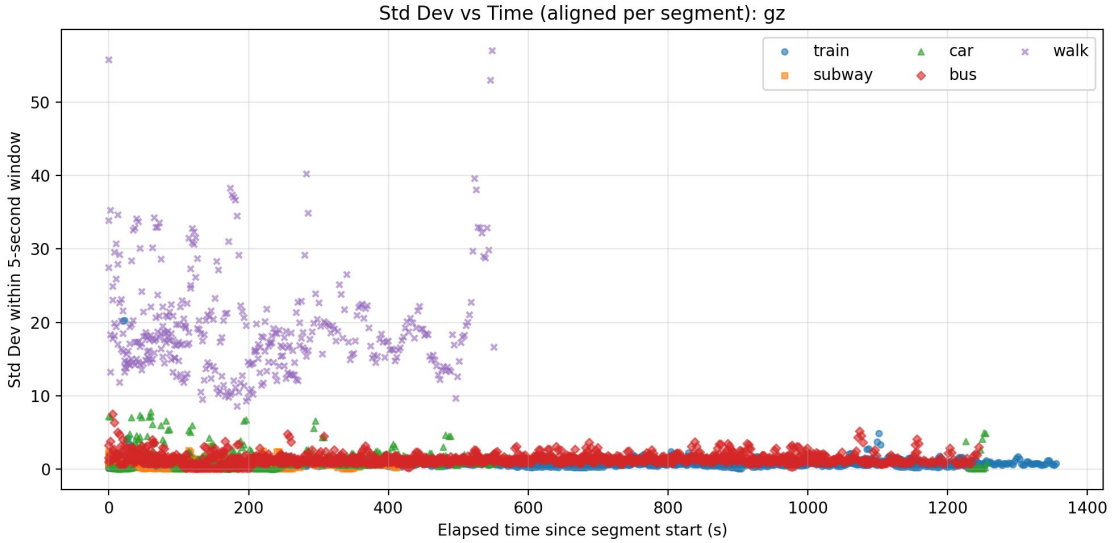
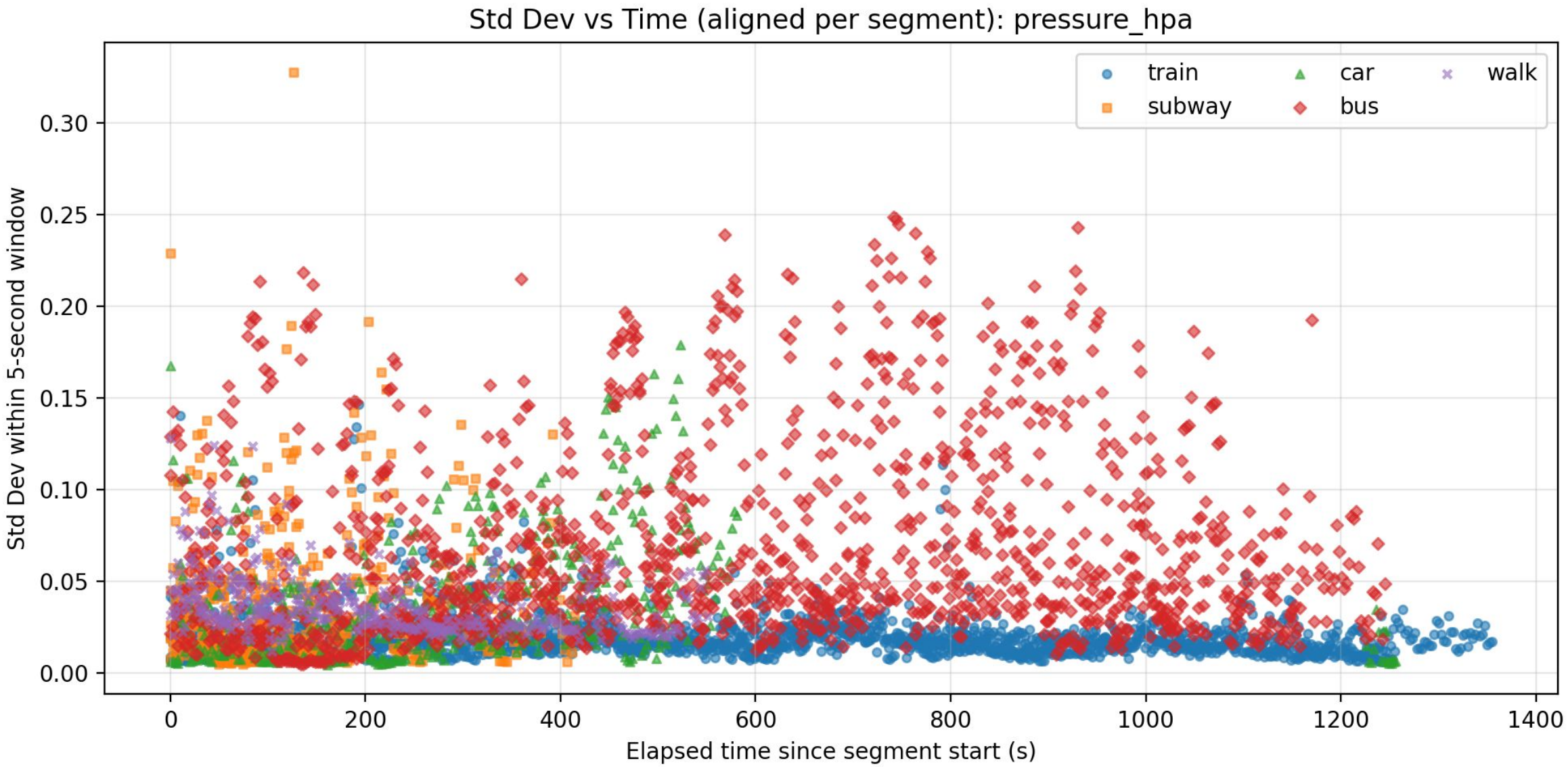




Appendix - Gyroscope Graphs







Appendix - Pressure Graph